Real-world Ethics for Self-Driving Cars

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ABSTRACT

Ethical and social problems of the emerging technology of self-driving cars can best be addressed through an applied engineering ethical approach. However, currently social and ethical problems are typically being presented in terms of an idealized unsolvable decision-making problem, the so-called Trolley Problem. Instead, we propose that ethical analysis should focus on the study of ethics of complex real-world engineering problems. As software plays a crucial role in the control of self-driving cars, software engineering solutions should handle actual ethical and social considerations. We take a closer look at the regulative instruments, standards, design, and implementations of components, systems, and services and we present practical social and ethical challenges that must be met in the ecology of the socio-technological system of self-driving cars which implies novel expectations for software engineering in the automotive industry.

KEYWORDS

Self-Driving Cars, Autonomous Cars, Trolley Problem, Decision Making, Ethics, Social Aspects, Software Engineering, Challenges

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1 PROBLEM

Self-driving cars have become part of the intense development all over the world, and they are attracting big societal attention as a revolution of transport systems [6, 17] that will affect society in fundamental ways. The interest of the general public up to now was mostly focused on the fact that decision-making is performed by a non-human agent [10], presented as a key issue for self-driving cars [3]. The questions discussed were of the kind: which person will an autonomous car choose to kill: a young lady or an old gentleman?

However, when it comes to societal and ethical aspects, self-driving cars are much more complex technology than what is reflected in the popular debate. We have the typical "policy-vacuum" problem [16] of computer ethics, which arises in situations for which we lack policies, experience, ethics, and laws. The Trolley Problem focus is fundamentally misguided and it is setting the wrong emphasis on a hypothetical unsolvable ethical dilemma instead of relevant challenges for real-world self-driving cars.

We highlight the following aspects that demonstrate the irrelevance of the Trolley Problem for the real-world ethics of self-driving cars:

(1) Currently, there are no fully autonomous cars and humans are still responsible for the decision making. The process of increasing automation goes via step-wise improved driving performance of a car based on machine learning. Two recent studies that compare crash experiences of automated vs. conventional vehicles show that automated vehicles perform better [2, 19]. The behaviour of the car is based on learning from experience and not on programming predefined hypothetical scenarios.
(2) The assumption made in the Trolley Problem about the deterministic nature of all the involved processes is wrong. It means that all the objects have perfectly known positions from which only one perfectly calculable consequence will follow. In the real world, we have a complex system under uncertainty that is not possible to predict exactly in real-time, and the way those phenomena are handled is by machine learning. Learning from real-world driving experiences leads to improved data, which constantly improve the capabilities of the self-driving cars.
(3) The Moral Machine experiment [1, 3], asking people all over the world about what they would do in a Trolley Problem situation, is often mentioned in connection to autonomous cars, in spite of the fact that it is about people and not about self-driving cars. It is not a way to understand what cars should do, as humans are known to be the main cause of car accidents. According to The National Highway Traffic Safety Administration, 94% of serious crashes are due to human errors [18].

We argue that, instead of Trolley Problem scenarios exploring differences among people in what they believe they would do in certain traffic situations, the relevance for the development of autonomous cars lies in the techno-social and ethical aspects of real-world engineering.

2 RESEARCH APPROACH

We use a hybrid interdisciplinary methodology in order to identify relevant ethical and societal challenges for the development of self-driving cars. It builds on insights from ethics theory with the significance of "policy vacuums" that are being filled by adding the most important aspects of emerging technologies. The combination of literature on value-based design and regulations, guidelines or standards [7–9, 11, 12, 15, 20] with the technical characteristics of

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present-day automated cars and their anticipated developments, allows extracting a list of most important topics, relevant for the work towards autonomous cars as a technology for good. This list provided a basis for discussions with experts, stakeholders and for seminars to validate our findings.

Problems identified in our study as relevant with respect to ethical theory and existing guidelines, standards, and codes of ethics, are meant to be used to orient technological approaches. We suggest focusing on the number of real-world problems presented in the following section. We also indicate in what way software engineering solutions are affected by the new focus on pragmatic aspects of ethics for self-driving cars.

3 RESULTS

After the misguided discussion trying to apply Trolley Problem, it is time to start addressing real-world ethical challenges and social consequences of self-driving vehicles [6], as this new technology is tested and gradually allowed on the roads under controlled conditions. The true ethical challenges include characteristics of the techno-social system supporting new technology, with the emphasis on maximising learning, on machine-, individual-, and social-level [4, 5]. We have identified technical challenges and their manifestations with respect to the following requirements: safety, security, privacy, trust, transparency, reliability, responsibility, and accountability, quality assurance, and sustainability. With regard to social challenges, we considered manifestations grouped by the following requirements: social challenges of disruptive technology, stakeholders -general public interests, and legislation, norms, policies, and standards. A detailed summary is presented in [14].

As an example of a technical challenge, we can mention the decision-making process (that the Trolley Problem focuses on) and its implementation, which is central for the behaviour of a car, and which might use unreliable or insecure technological components. To mitigate these problems, self-driving cars technology should follow new laws, ethical principles, and guidelines. This presents new expectations, which affect software engineering that is involved in all its stages - from its regulatory infrastructure to the requirements engineering, development, implementation, testing, and verification [4, 13]. As software is an integral part of a complex software-hardware-human-society system, we presented different types of issues that we anticipate will affect software engineering in the near future. It is of key importance to include ethical thinking and reasoning into the design and development process in every phase, from requirements, till testing, maintenance, and evolution. Architectural and design decisions should be taken through a process that includes ethics as a first-class actor (required non-functional property) and by involving stakeholders that are relevant to this concern. These architectural and design decisions should then be embedded into the code that will run the self-driving vehicles and ensure its ethicality. It is also necessary to enforce the transparency on those processes and software, to enable independent evaluations. Proper development processes, supported by suitable organisation structure should promote and enable a serious discussion of ethics, and emphasise the human values, to make sure that the freedom of choice does not disappear in the new era of fully autonomous self-driving vehicles.

Finally, we want to point out the importance of the ecology of the whole socio-technological system, where ethics is ensured through education, constant information, and negotiation of priorities in the value systems, as well as constant learning process within technology and its social environment. In the development process values and ethics come first, followed by legislation and standardisation which is constantly monitored in practice and validated with value- and ethical standards. That is why we emphasise the central role of real-life ethics as a basis that will sustain and inform ethically sound emerging technology.

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